Formula 1 Podium Prediction

CSEN 903 Advanced Computer Lab — Milestone 1 analytical report

Team	Team 40 — [member names and IDs]
Kaggle notebook	[link]
GitHub repository	[link]
Data	Formula 1 World Championship 1950–2024 (Kaggle / Ergast), 14 relational tables

Summary. We built a validated, leakage-safe modelling table with one row per driver per race (26,668 rows, 1950–2024) and predicted whether a driver finishes on the podium using only information available after qualifying. Trained on 2003–2019, tuned on 2020–2021 and tested once on 2022–2024, all four required models beat the grid-position baselines. The final model — a bonus race-aware ensemble — reaches on the unseen test seasons:

0.717	0.939	0.683	0.617	0.765
PR-AUC (primary)	ROC-AUC	F1	Precision	Recall

versus PR-AUC 0.647 / F1 0.649 for the rule "a driver starting in the top 3 finishes on the podium".

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1. Task and leakage rules

- **Unit of analysis:** one driver entering one race; target `podium = positionOrder ≤ 3`.
- **Prediction cutoff:** after qualifying, before the start. Allowed: this race's grid and qualifying position, and anything computed from *earlier* races. Not allowed: this race's result, points, laps, status, fastest lap, lap times, pit stops, and the standings row of this race (it already contains the result).
- **Time order:** always season then round. `raceId` is *not* chronological (2009 has ids 1–17, 1950 has 833–839).

2. Data audit and cleaning

The 14 raw CSVs are read once and never modified; every step works on copies. Each cleaning step answers a problem found in the exploratory analysis before cleaning.

Finding (EDA before cleaning)	Evidence	Action
Missing values written as the text `\N`; numeric columns read as text	160,076 cells, 77 % in `results`	`\N` → NaN; a column becomes numeric only if every non-missing value is a number (10 columns)
Durations and dates stored as text (`'1:27.452'`)	q1–q3, fastest lap, pit and lap times	Converted to seconds / dates; checked against the millisecond columns (difference 0.0 s)
Grain broken: 91 extra (raceId, driverId) rows	85 pairs, 1950s shared drives and double entries	Keep the row with the best `positionOrder`; all 22 podiums in duplicated groups preserved → 26,668 rows
Fastest-lap rank 0 is not a rank	258 rows, no fastest-lap time, median 0 laps	Set to NaN
`grid = 0` is not a slot (failed to qualify, pit-lane start)	peak 13 % of rows in the 1980s	Flag starters; in features moved behind the last car (`grid_pos`)
Pit stops of minutes (red flags count as stops)	517 stops > 60 s (4.5 %), up to 51 min	Flagged; excluded from the pit-time average
Inconsistent names used in joins	`'Argentinian '` / Argentine; USA / United States	Stripped and unified
Primary and foreign keys	26 reference checks	All ids unique and non-null; 0 orphan foreign keys before and after cleaning

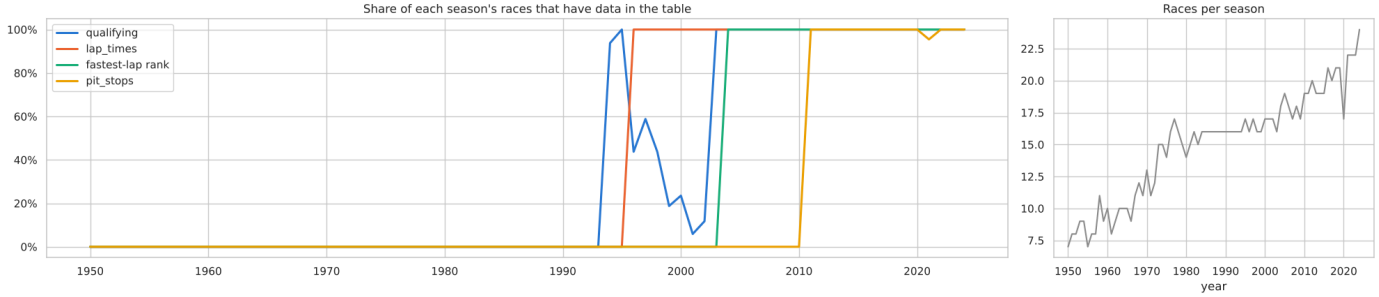


Figure 1. Share of each season's races with data in the session tables. Qualifying is complete only from 2003 (gaps 1996–2002), lap times from 1996, fastest-lap rank from 2004, pit stops from 2011. These dates decide where engineered features are real and where they hold fill values.



Figure 2. Duplicate driver-race rows are a 1950s phenomenon (shared drives).

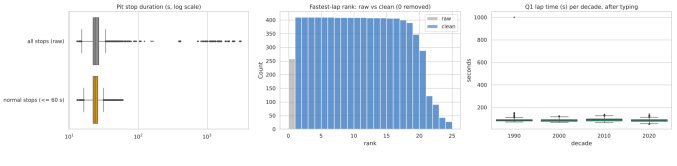


Figure 3. After cleaning: pit-stop outliers removed from the average, rank 0 gone, qualifying times numeric.

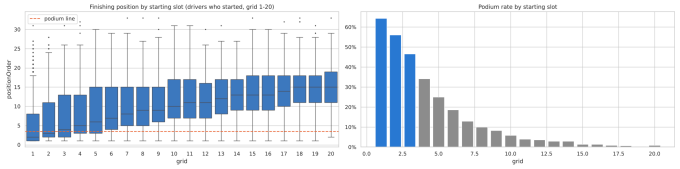


Figure 4. Finishing position and podium rate by grid slot: pole → 64 %, P5 → 25 %, P8 → 10 %.

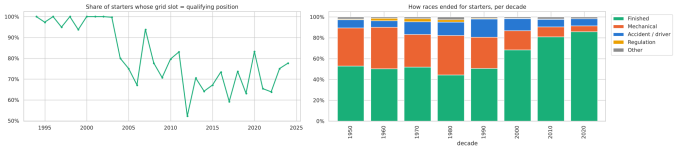


Figure 5. Grid ≠ qualifying for 20–37 % of modern starters (penalties); mechanical failures fell from 32–40 % of starts (1950s–80s) to 6 % (2020s).

3. Data-engineering questions

All answers use the cleaned grain (one row per driver per race), only drivers who actually started, and exclude the Indianapolis 500 (1950–60, a different series' drivers on an oval).

Q1. Which circuits convert a front-row start into a podium most often, and does the ranking change after 2014?

Joins: results → races → circuits, and results → qualifying. Front row = slot 1–2, measured by **grid** (after penalties) and by **qualifying position**. Eras 1994–2013 vs 2014–2024 (qualifying data starts in 1994); ≥ 10 front-row starts per era.

- **2014–2024 (grid):** Monaco 89 %, Suzuka 89 %, Yas Marina 86 % convert most often; Hungaroring least (59 %).
- **The ranking changes almost completely:** rank correlation between eras −0.04 (grid) and 0.14 (qualifying). Hungaroring went from 2nd (72 %) to last; only Suzuka stays at the top.
- **The level rose:** 65 % → 76 % (grid), 65 % → 74 % (qualifying) — hybrid-era dominance and better reliability.
- **Grid vs qualifying:** of 983 front-row starts with qualifying data, 36 qualifiers lost the front row to a penalty and 36 drivers moved onto it.

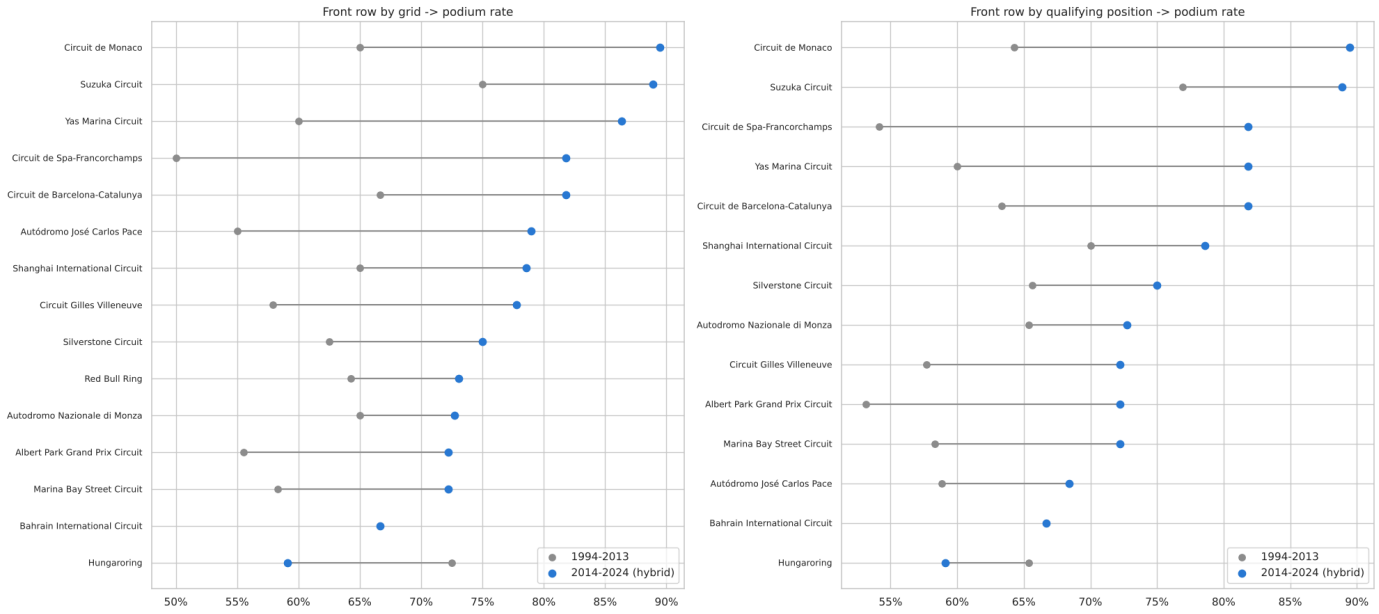


Figure 6. Front-row → podium conversion per circuit, by grid (left) and by qualifying position (right).

Use for the model: even from the front row a podium is not certain (59–89 %), so more than the grid is needed; grid and qualifying are both kept (with their difference); circuit rankings are unstable across eras, so no fixed circuit-conversion feature was added.

Q2. Do drivers podium more often at home, after controlling for starting position?

Joins: results → races → circuits (country) and results → drivers (nationality → country map). Home and away drivers are compared **on the same grid slot**; differences are averaged with home-start weights; 95 % interval by resampling whole races (1,000×).

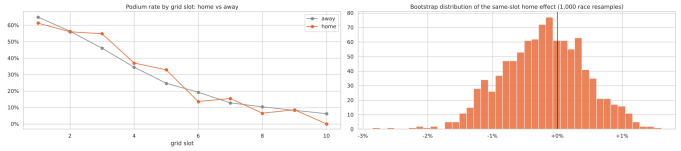


Figure 7. Podium rate per grid slot, home vs away, and the bootstrap distribution of the adjusted effect.

- Raw: home 11.9 % vs away 13.9 % — but home drivers also start further back (12.7 vs 11.7).
- **Same grid slot: −0.2 percentage points (95 % interval −1.4 to +1.0).** The interval contains 0.
- **Answer: no home advantage** once the start is controlled for.
- **Use:** `is_home_race` is dropped from the model features.

Q3. Which constructors have the highest mechanical-retirement rate in 2014–2021 vs 2022–2024?

Joins: results → status (each of the 139 status texts classified; "mechanical" = a car component failed; crashes, driver errors, regulation decisions, tyres/wings and unknown "Retired" excluded) → races → constructors. Rate = mechanical retirements / race starts; ≥ 40 starts per period.

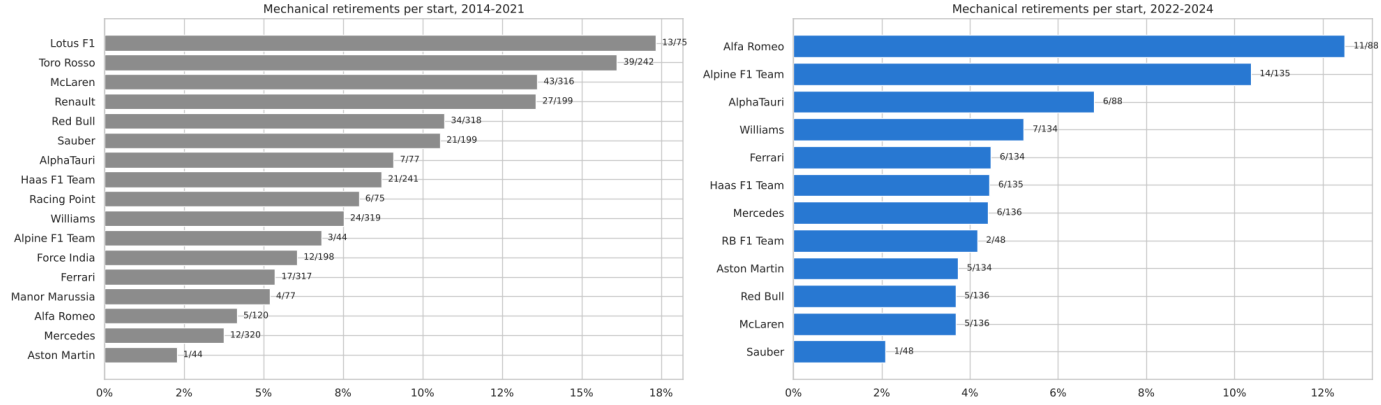


Figure 8. Mechanical retirements per start (labels: failures / starts).

- **2014–2021:** Lotus F1 17 % (13/75), Toro Rosso 16 %, McLaren 14 % (its Honda power-unit years) — Renault equal.
- **2022–2024:** Alfa Romeo 12.5 % (11/88), Alpine 10 %, AlphaTauri 7 %.
- Overall reliability improved from 9.2 % to 5.5 % of starts; McLaren went from one of the worst to 3.7 %; Mercedes was the most reliable large team in 2014–2021 (3.8 %).
- **Use:** about 1 in 11 starts ended with a broken car — uncertainty no pre-race feature can remove; it reappears as the main cause of false positives (Section 8).

Bonus questions: testing the logic behind our engineered features

Question (joins)	Result	Verdict
B1. Does recent form add information beyond the starting slot? (results, races, rolling form)	From the same front row, best-form third 67 % vs worst third 41 %; same pattern in every grid band	Confirmed
B2. Does the faster qualifier also beat his teammate in the race? (results, qualifying, status)	70 % of 4,424 duels where both finished (50 % = no information)	Confirmed
B3. After a grid penalty, does grid or qualifying decide? (results, qualifying)	Penalised top-3 qualifiers podium 32 % vs 64 %; but from the same slot they podium +3.7 points more (95 % CI +1.6 to +5.5)	Both needed
B4. Does track history matter beyond career level? (results, races/circuits)	No clear difference within the same career level (e.g. 36 % vs 36 %)	Not confirmed (reported as negative result)
B5. After a retirement, is running position a better form signal than finishing position? (results, status, lap_times)	Previous finishing position r = −0.04; 5-race running position r = −0.42	Confirmed

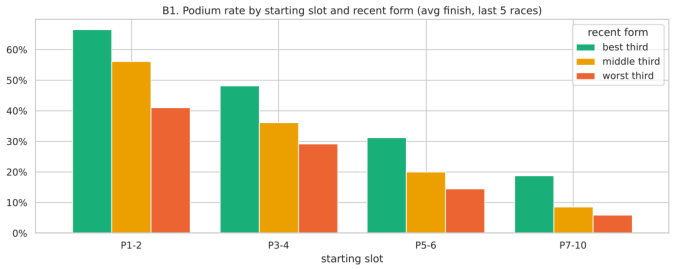


Figure 9. B1: podium rate by starting slot and recent form.

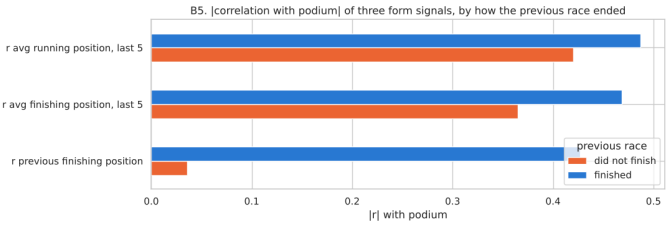


Figure 10. B5: strength of three form signals after a finished vs unfinished previous race.

4. Feature engineering and selection

Leakage-safe construction. Inside each driver (or team), sorted by season and round, every history feature uses `groupby(...).transform(lambda s: s.shift(1).rolling(5).mean())` — the shift *and* the window stay inside the group, so only earlier races enter. Standings are taken after the *previous* round of the same season. A hand recomputation of 200 random rows from strictly earlier races matched 200/200. Features are built on all seasons first, then filtered (Section 7, P2).

Feature (21 selected)	Built from	Known before the start because
`quali_position`, `grid_pos` (grid 0 → behind last car), `grid_penalty_delta`	qualifying, results.grid	qualifying and the grid are set before the race
`outqualified_teammate` (1 / 0 / 0.5 no comparison)	qualifying within the team	qualifying order
`driver_prev_position`, `driver_prev_podium`, `driver_avg_pos_last_5`	results + races	earlier races (shift 1)
`driver_avg_lap_pos_last_5`, `driver_avg_speed_rank_last_5`	lap_times, results.rank	earlier races
`driver_career_avg_pos`, `driver_career_podiums`, `driver_career_races`, `driver_avg_pos_at_circuit`	results + races (+ circuitId)	expanding over earlier races
`pre_race_wins`, `won_previous_race`, `points_share` (points ÷ leader's points)	driver_standings + races	standings after the previous round
`constructor_prev_pos`, `constructor_avg_pos_last_5`, `constructor_career_podiums`	results grouped by team	earlier races
`pre_race_constructor_wins`, `constructor_points_share`	constructor_standings + races	standings after the previous round

Fill rules (pre-race facts only): no earlier race → 0 for counts/standings and 20 (back of field) for positions; first visit to a circuit → career average; missing qualifying → grid slot; tables that do not exist yet (lap times < 1996, fastest-lap rank < 2004) → 20.

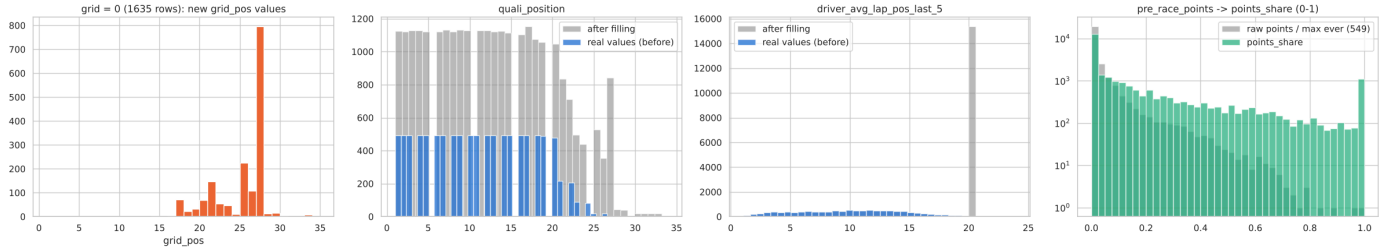


Figure 11. Effect of preprocessing on the distributions: grid 0 moved behind the field, filled qualifying, the fill-value spike of lap position, raw points → 0–1 share.

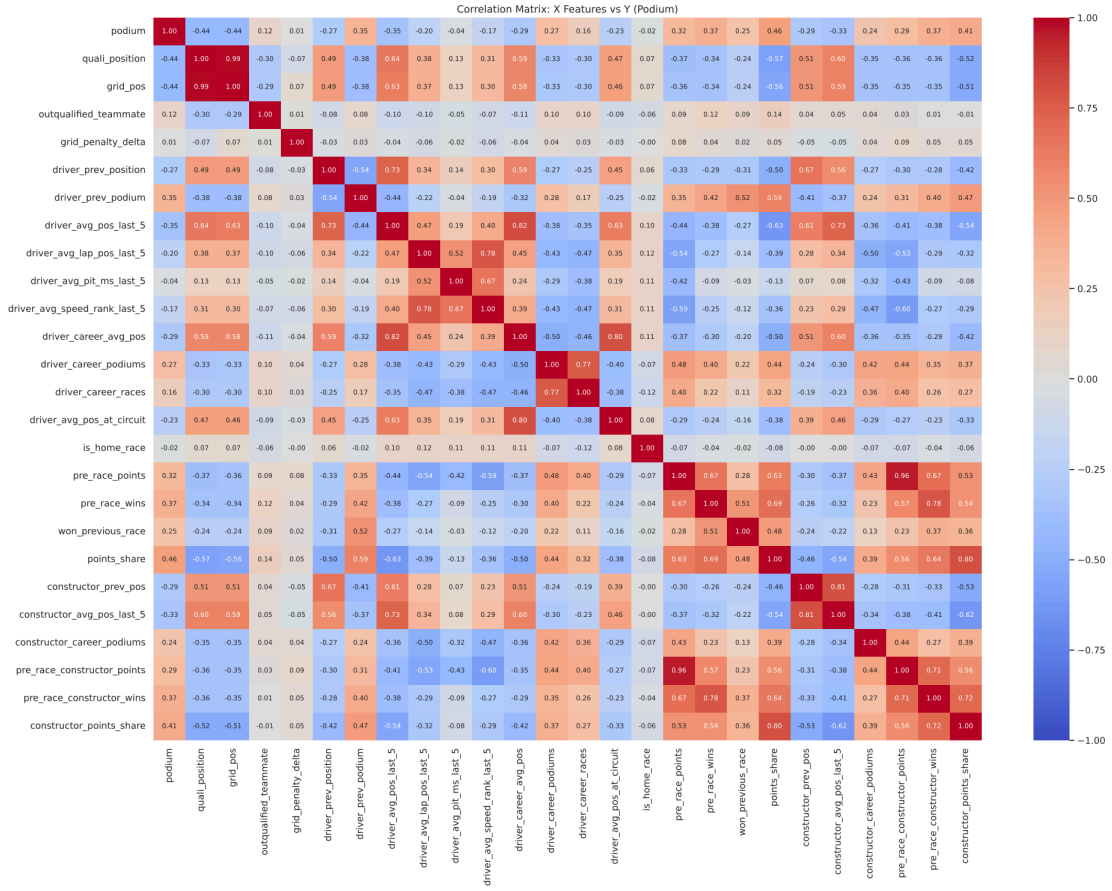


Figure 12. Correlation matrix of all candidate features and the target (all seasons).

Reading the matrix. `points_share` has the highest correlation with podium (+0.46): it summarises the whole season so far for driver and car on a 0–1 scale that is comparable across points systems (raw points: +0.32). `quali_position` / `grid_pos` follow (−0.44) — and are the strongest by rank correlation, because their effect is steep and non-linear (P1 64 %, below 10 % from P9). `outqualified_teammate` is weak alone (+0.12) but three times as strongly linked to the grid (−0.30), and its value depends on the car: in top teams beating the teammate raises the podium rate from 29 % to 58 %, in back-marker teams from 1.4 % to 3.1 %. Several correlations are **era artefacts** of fill values: lap position is −0.20 on all rows but −0.48 on rows with real values.

Dropped: `driver_avg_pit_ms_last_5` (a fill value for 78 % of rows, weakest form signal even on real values), `is_home_race` (no effect, Q2), raw `pre_race_points` / `pre_race_constructor_points` (replaced by the shares; 0.96 with each other), raw `grid` (replaced by `grid_pos`).

5. Split, metrics and pre-training checks

- **Split by season** (whole races stay together; the model always predicts seasons after the ones it learned from): train ≤ 2019, **validation 2020–2021** (39 races, 780 rows), **test 2022–2024** (68 races, 1,359 rows). A random split would mix races of the same season across sets and leak the future.
- **Primary metric: PR-AUC** — podiums are only ~15 % of rows, and PR-AUC focuses on ranking the rare positive class. **Secondary:** ROC-AUC (overall ranking) and F1 at a threshold, with precision and recall. Accuracy is misleading (always "no podium" ≈ 85 %).
- Hyperparameters, the F1-optimal threshold and every choice are made on train/validation only; test is scored once.

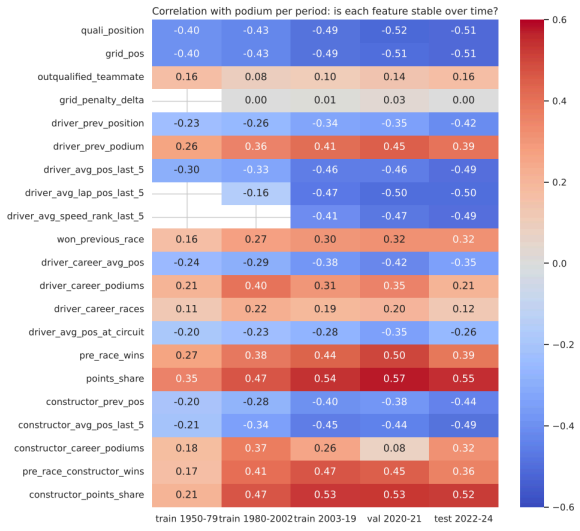


Figure 13. Correlation with podium per period: same direction in every era, stronger in modern seasons; 2003–19 resembles validation and test.

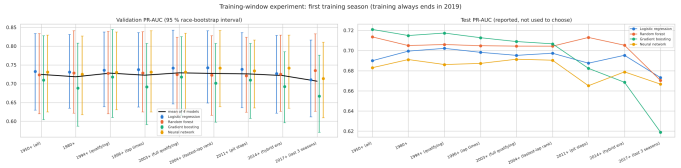


Figure 14. Training-window experiment (year ablation): validation PR-AUC with race-bootstrap intervals, and test.

Window choice. Pre-stated rule: highest mean validation PR-AUC of the four models → **2003+** (0.729; 1994+ 0.728, 1950+ 0.724, 2017+ 0.707). Windows are close; very short windows are clearly worse. 2003 is also the first season with qualifying data for every race. Training set: 2003–2019, 321 races, 6,831 rows.

6. Models, baselines and results

Model	How it works	Tuned on validation	Limitation
Logistic regression	weighted sum of standardised features → sigmoid	C = 0.1	straight-line effects only
Random forest	300 trees on bootstrap samples, averaged	min leaf 50, 50 % features per split	smoothed probabilities, no extrapolation
Gradient boosting	small trees added sequentially, each fixing previous errors	31 leaves, 100 trees, lr 0.03, L2	overfits fastest (train 0.79 vs val 0.73)
Shallow neural network (Keras)	1 hidden layer of 16 ReLU units, sigmoid output, Adam, binary cross-entropy, balanced class weights, early stopping	12 architectures compared	needs scaling, sensitive to random start, over-predicts probabilities (class weights)

Model	thr.	PR-AUC train / val / **test**	ROC-AUC val / **test**	F1 val / **test**	Precision **test**	Recall **test**
Baseline: grid ≤ 3 rule	0.333	0.626 / 0.639 / **0.647**	0.909 / **0.914**	0.692 / **0.649**	0.650	0.647
Baseline: logistic regression (grid + qualifying)	0.337	0.631 / 0.662 / **0.652**	0.920 / **0.919**	0.696 / **0.672**	0.600	0.765
Logistic regression	0.369	0.685 / 0.743 / **0.696**	0.928 / **0.935**	0.715 / **0.671**	0.624	0.725
Random forest	0.372	0.746 / 0.733 / **0.707**	0.935 / **0.939**	0.709 / **0.675**	0.650	0.701
Gradient boosting	0.374	0.793 / 0.729 / **0.709**	0.932 / **0.937**	0.704 / **0.662**	0.644	0.681
Shallow neural network	0.737	0.729 / 0.751 / **0.677**	0.931 / **0.929**	0.718 / **0.661**	0.617	0.711
Final: race-aware tier (bonus)	**0.403**	**0.761 / 0.752 / 0.717**	**0.933 / 0.939**	**0.715 / 0.683**	**0.617**	**0.765**

- **All models beat both baselines** on PR-AUC, ROC-AUC and F1. The gain over the grid rule is moderate (+0.05–0.06 PR-AUC): the starting slot carries most of the information.
- The required models are within a few hundredths of each other and their order changes between validation and test — no single model dominates. Test scores are lower than validation (new 2022 rules; an unusually open 2024).
- Accuracy is 0.89–0.90 for every model *and* baseline, which is why it is not used to compare models.

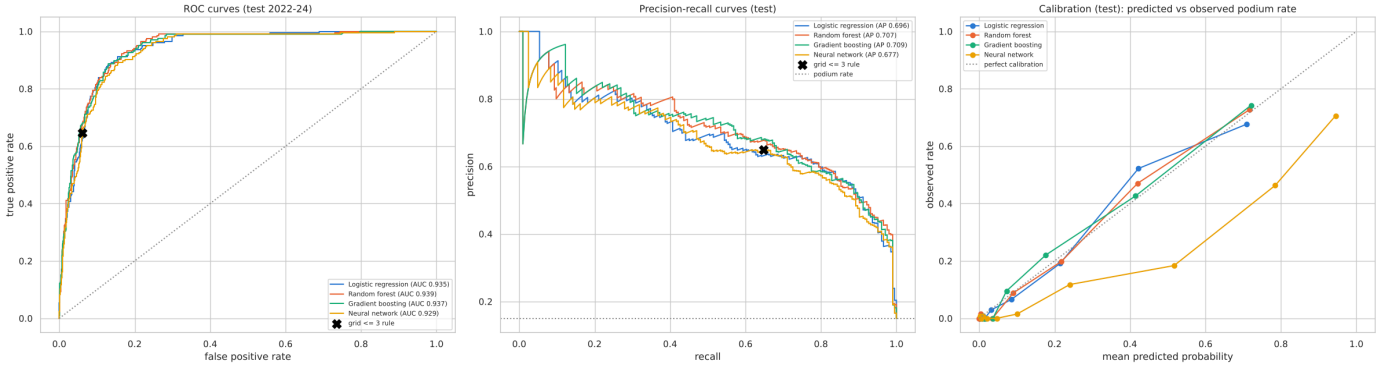


Figure 15. Test ROC and precision-recall curves (× = grid rule) and calibration. Tree models and logistic regression are well calibrated; the class-weighted network over-predicts.

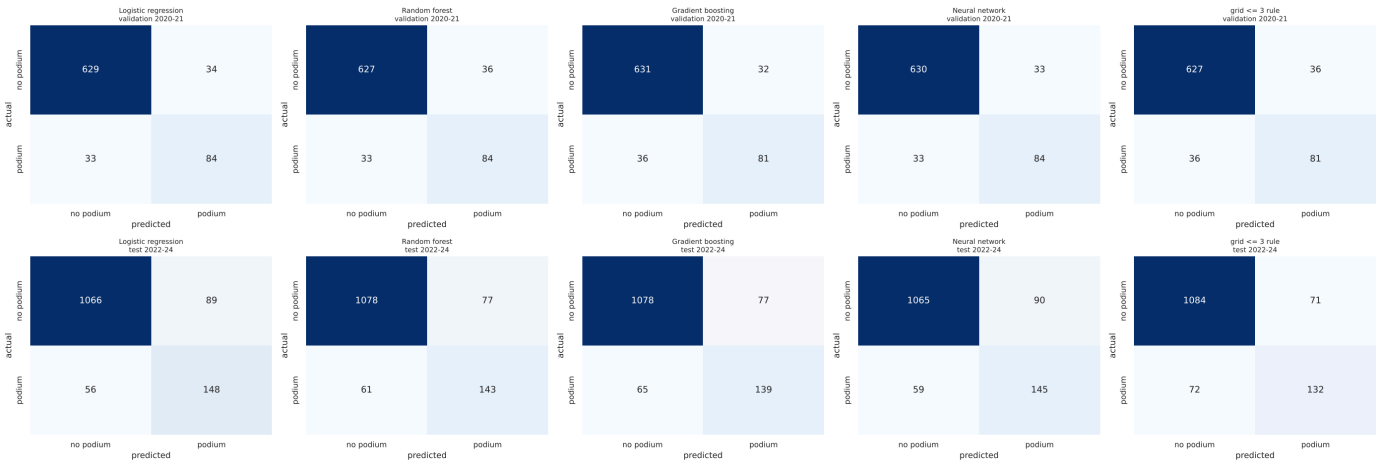


Figure 16. Confusion matrices at the validation threshold (top: validation, bottom: test). Models find 139–148 of 204 test podiums vs 132 for the rule.

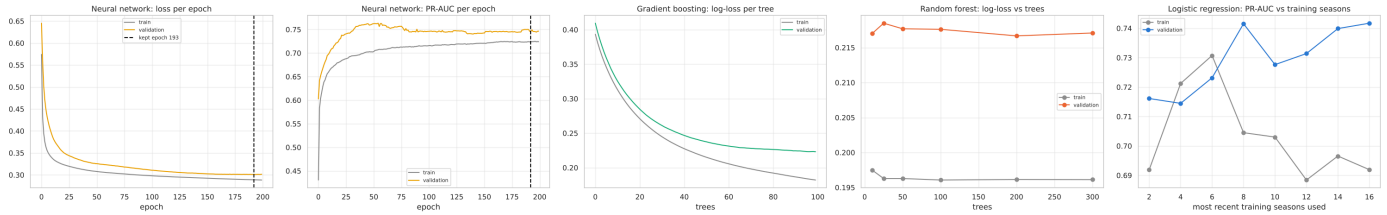


Figure 17. Training vs validation curves: network loss and PR-AUC per epoch (kept epoch marked), gradient-boosting log-loss per tree (validation flattens after ~70 trees), random forest vs trees, logistic regression vs number of training seasons.

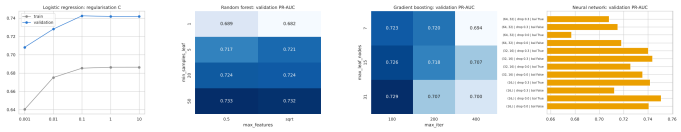


Figure 18. Hyperparameter tuning on validation PR-AUC.

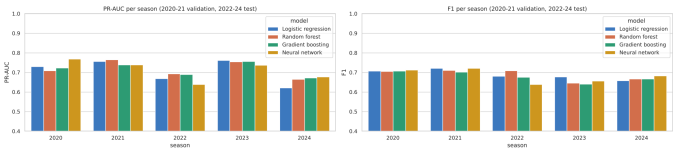


Figure 19. PR-AUC and F1 per season: 2023 easiest (Red Bull won 21/22), 2024 hardest (four winning teams).

7. Experiments: preprocessing order and ablations

Preprocessing-order experiments (logistic regression and gradient boosting, tuned, 2003+)

Experiment	Variant	Val PR-AUC (LR / GB)	Test PR-AUC (LR / GB)	Conclusion
P1: order of shift and rolling window	A. shift → window, inside each driver (ours)	0.743 / 0.729	0.696 / 0.709	B looks better only because it contains this race's result — the typical leakage alarm. C silently flips `driver_career_podiums` to r = −0.06 (our first bug).
	B. no shift (leaky)	0.801 / 0.806	0.761 / 0.803	
	C. window applied after the grouping ended	0.739 / 0.707	0.698 / 0.710	
P2: build features then filter vs filter first	features on all seasons, then filter (ours)	0.743 / 0.729	0.696 / 0.709	Filtering first resets every career at 2003 (champions look like rookies); consistently slightly worse.
	filter to 2003+ first	0.742 / 0.718	0.689 / 0.708	
P3: handling placeholders	(a) back-of-field fill (ours)	0.743 / 0.729	0.696 / 0.709	≤ 0.01 differences: only 1–6 % of 2003+ rows hold placeholders, so the window choice already solved it.
	(b) kept missing (GB native, LR train median after split)	0.740 / 0.719	0.696 / 0.715	
	(c) fill + missing indicators	0.739 / 0.729	0.696 / 0.709	

Feature-group ablation (all four models, race-bootstrap intervals)

Pre-stated rule: a group helps if removing it lowers validation PR-AUC with the 95 % interval below 0 for at least two models.

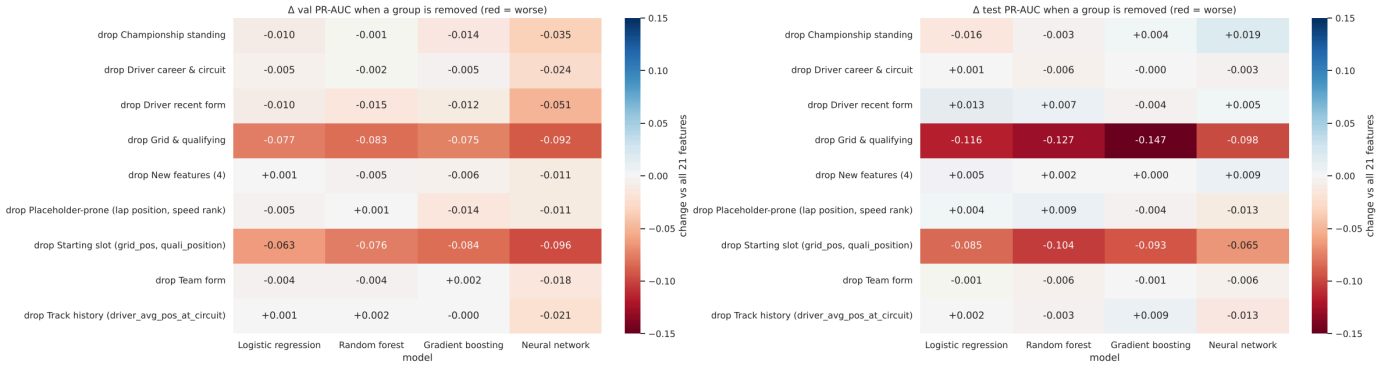


Figure 20. Change in validation (left) and test (right) PR-AUC when a group is removed.

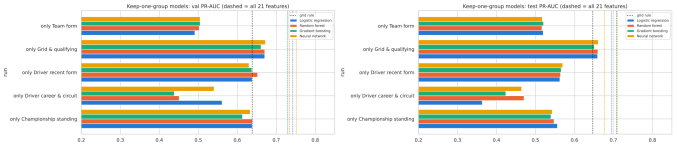


Figure 21. Keep-one-group models (dashed: all 21 features; dotted: grid rule).

- **Only grid & qualifying passes the rule** (all 4 models): −0.075 to −0.092 validation, −0.10 to −0.15 test.
- Every other group removed alone changes the statistical models by < 0.015 — **they overlap** (team form, standings and form all describe current car/driver strength).
- Keep-one: grid & qualifying alone 0.66–0.67; recent form 0.63–0.65; standings 0.61–0.64; team form ~0.50; career 0.44–0.56. **All together 0.73–0.75** (+0.07–0.08).
- Our 4 new features and track history (B4) show no clear contribution.

Error-driven ablation

From the SHAP analysis of validation errors (Section 8) the three features most often responsible for an error were removed (`quali_position`, then `grid_pos`, then lap position). Removing qualifying alone changes little; removing qualifying and grid lowers validation PR-AUC by 0.06–0.10 and missed podiums do not go down (they rise for three of four models). The features that "cause" errors are the same that make the correct predictions — errors come from race events, so the full set is kept.

8. Explainability (XAI)

> **Not causal.** All explanations show what the models *rely on* to predict, not what *causes* a podium. A feature can be important because it is correlated with the real cause (car speed, driver skill, luck), and correlated features can share importance arbitrarily.

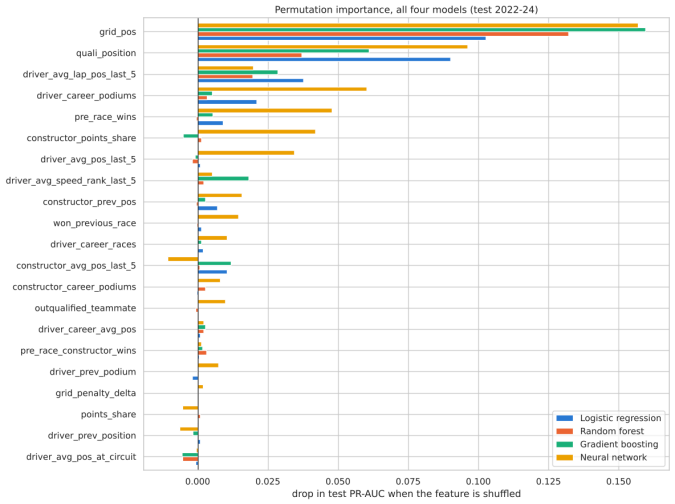


Figure 22. Permutation importance, all four models (drop in test PR-AUC when shuffled).

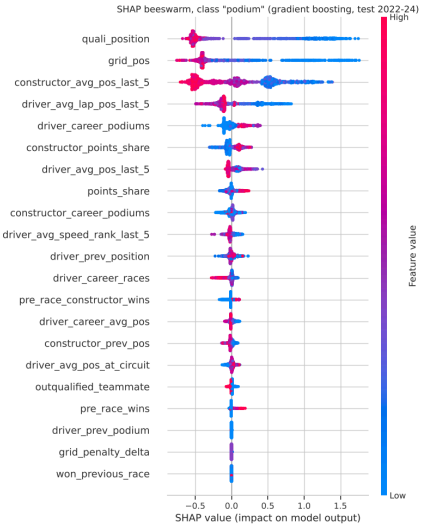


Figure 23. SHAP beeswarm, gradient boosting, class "podium" (the "no podium" plot is its mirror image).

- **Global:** `grid_pos` is the most important feature for every model (0.10–0.16), then `quali_position`; SHAP adds team form (`constructor_avg_pos_last_5`) and running position. The network spreads importance more widely.
- **Shape (SHAP dependence):** qualifying P1–P6 pushes up by 1–1.6 log-odds, the push collapses at P7–P8 and is flat from P9 — the same step seen in the pre-training EDA. `outqualified_teammate` is tiny and slightly negative *given* the qualifying position (being ahead of a teammate at a given position means the teammate is further back — a weaker car): a conditional effect can differ from the marginal correlation.

SHAP global vs the correlation matrix before training

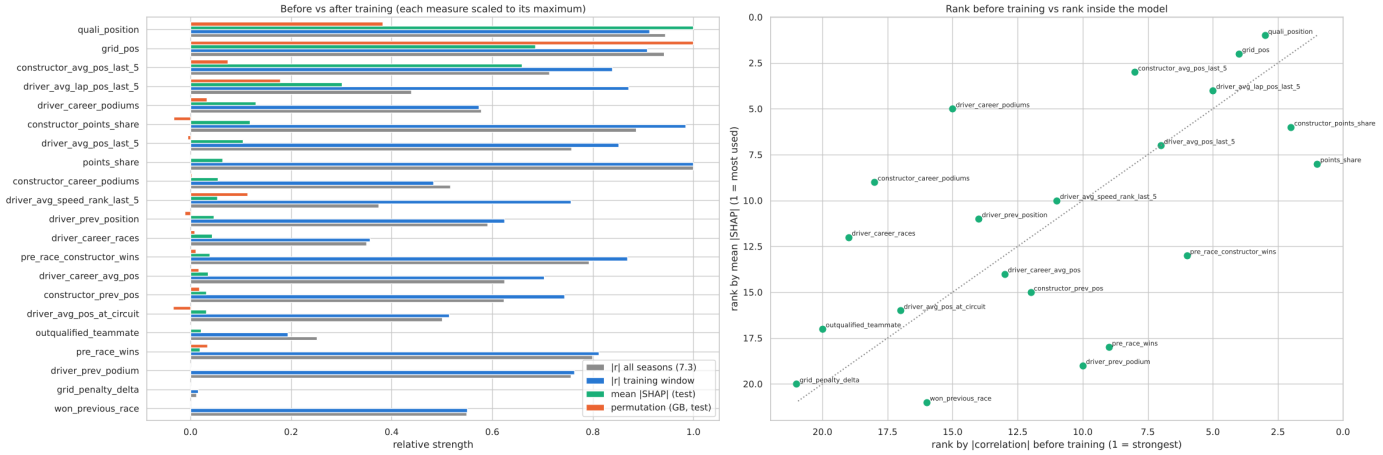


Figure 24. Correlation before training vs mean |SHAP| and permutation importance after training; right: rank vs rank.

Rank agreement between mean |SHAP| and |correlation| is **0.41** (all seasons) and **0.61** (training seasons). Both put the starting position and team/recent form near the top. `points_share` is #1 by correlation but #8 by SHAP and `pre_race_wins` / `driver_prev_podium` fall to the bottom: correlation counts a feature's *total*

association, SHAP only what it adds inside the model, where grid and team form already explain it. Lap position is #17 in the all-season matrix but #4 by SHAP — the placeholder effect found before training. Correlation was a good first filter but cannot reveal redundancy.

Local explanations (SHAP waterfall + LIME on the same predictions, test seasons)

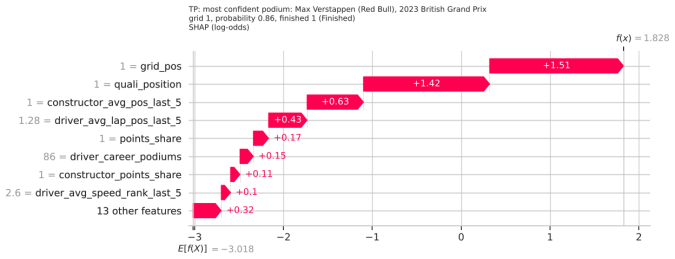


Figure 25. True positive: Verstappen, 2023 British GP (p = 0.86). Pole and P1 qualifying (+1.5/+1.4 log-odds), best team and running position.

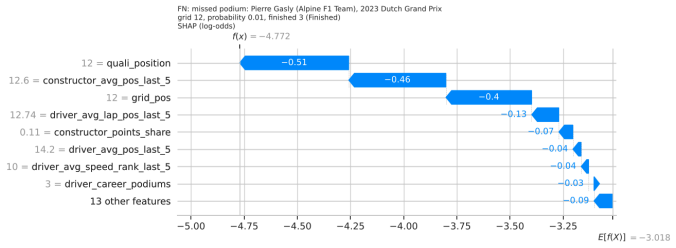


Figure 26. False negative: Gasly, 2023 Dutch GP (P12, p = 0.008) — a rain-affected race; grid, qualifying and Alpine's form push to ~0.

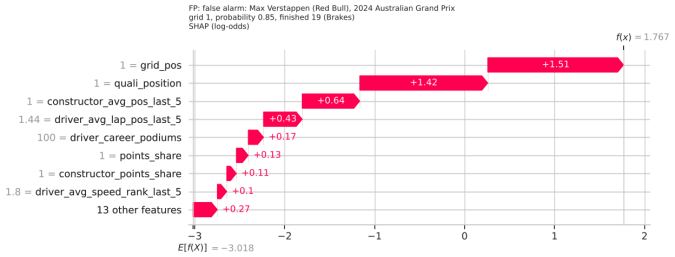


Figure 27. False positive: Verstappen, 2024 Australian GP (pole, p = 0.85) — retired with a brake failure.

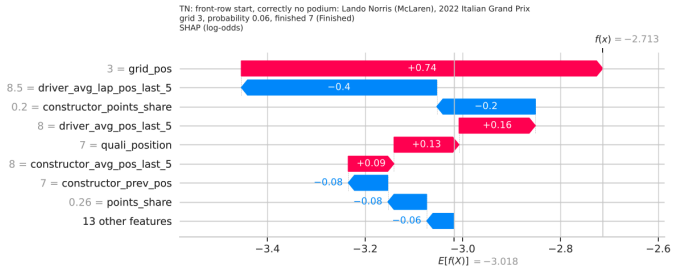


Figure 28. True negative: Norris, 2022 Italian GP — started P3 but qualified P7; the model trusts qualifying (p = 0.06); he finished 7th.

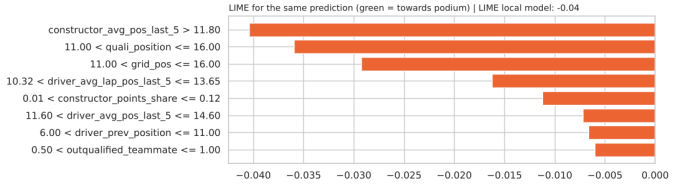


Figure 29. LIME for the Gasly case: the same features (team form, qualifying, grid) push towards "no podium".

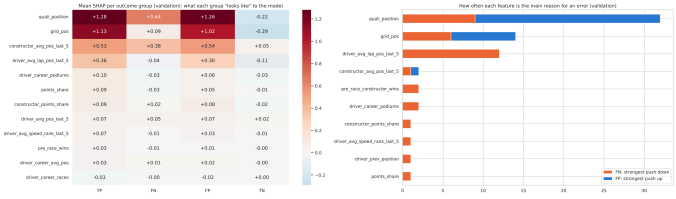


Figure 30. Validation error analysis: mean SHAP per outcome group and the feature that most often causes each error.

What the ten local cases show (all in the notebook, SHAP and LIME agree on the main features each time): missed podiums are drivers from P12–P15 in wet or chaotic races (Gasly 2023 Netherlands and 2024 Brazil, Pérez 2023 Austria); false alarms are front-row cars hit by race events (brake failure, collision, damage); **41 % of validation false positives did not finish**. The data contains no weather, safety-car or reliability-in-race information, which is the main source of remaining error.

9. Bonus: race-aware tier and final model

A podium is relative — exactly three per race. The bonus tier adds this structure, each step measured separately: (1) **race-relative features** — each driver's rank and gap to the best driver *in the same race* for team form, recent form, running position, career average and both points shares, plus field size (pre-race values of the same race, so leakage-safe); (2) an **ensemble** averaging the four tuned models; (3) **race normalisation** so probabilities sum to 3 per race.

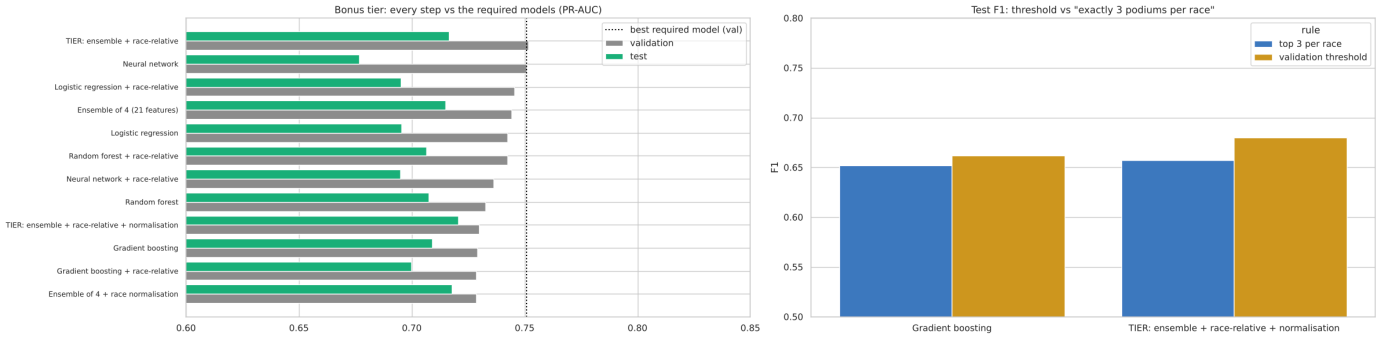


Figure 31. Every tier step vs the required models; right: threshold vs "exactly 3 podiums per race".

- The ensemble alone already beats every single model on test (0.715). **Ensemble + race-relative features has the best validation PR-AUC of all candidates (0.752)** and test 0.717 — above every required model on test (+0.008 over gradient boosting, +0.040 over the network).
- Honest margin: on validation it is only 0.001 above the tuned network, so the improvement is real on test but not large or guaranteed. Race normalisation raised test PR-AUC (0.721) but lowered validation, so by our rule it was not chosen; a fixed "top 3 per race" rule gave lower F1 than the validation threshold.
- **Final model** (chosen by the pre-stated rule: highest validation PR-AUC): the race-aware ensemble, 21 + 13 features.

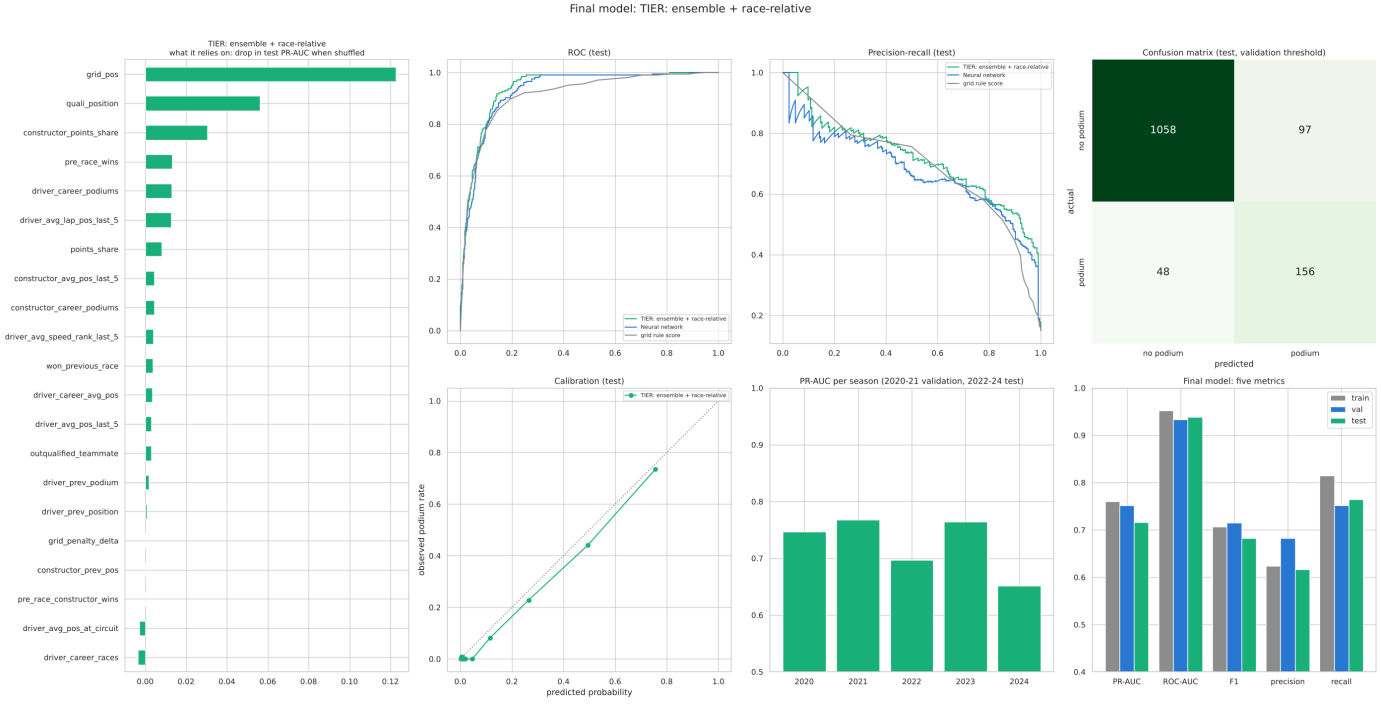


Figure 32. Final model: what it relies on (permutation through the full pipeline), ROC/PR vs best required model and grid rule, test confusion matrix (156 TP, 48 FN, 97 FP), calibration close to the diagonal, PR-AUC per season, five metrics on train/validation/test (0.761 / 0.752 / 0.717 PR-AUC: no strong overfitting).

All experiments at a glance (validation PR-AUC used for decisions; full chart of all 142 results in notebook Section 17.3)

Experiment	Val PR-AUC	Test PR-AUC	Note
P1-B leaky rolling (no shift), gradient boosting	0.806	0.803	invalid by construction — leakage alarm
Final: ensemble + race-relative features	**0.752**	**0.717**	**chosen by the pre-stated rule**
Tuned shallow neural network	0.751	0.677	best required model on validation
Logistic regression + race-relative features	0.746	0.695	
Ensemble of 4 (21 features)	0.744	0.715	
Ensemble + race normalisation (tier)	0.730	0.721	best test, lower validation → not chosen
Training window 2017+ (mean of 4 models)	0.707	0.657	too little data
Drop grid & qualifying (4 models)	0.650–0.666	0.562–0.580	the indispensable group
Grid ≤ 3 rule (baseline)	0.639	0.647	

10. Inference function

`predict_podium(entries)` takes the pre-race entries of one race (one row per driver; any feature may be missing), applies exactly the fill rules of Section 4, computes the race-relative features, and returns each driver's podium probability and the decision at the validation threshold (0.403).

2024 Abu Dhabi GP	Grid	Complete record	With missing values	Values filled	Actual
Lando Norris (McLaren)	1	0.832 → podium	0.525 → podium	6	Podium (won)
Max Verstappen (Red Bull)	4	0.687 → podium	0.687	0	No (6th)
Oscar Piastri (McLaren)	2	0.657 → podium	0.660	0	No (lap-1 collision)
Carlos Sainz (Ferrari)	3	0.596 → podium	0.595	0	Podium
Charles Leclerc (Ferrari)	19	0.164	0.168	0	Podium (from P19 after a penalty)
Debutant (only grid P10 + Mercedes team form)	10	—	0.143	15	—

With six values removed (qualifying, standings, recent form) Norris drops from 0.83 to 0.53 — the fills are deliberately conservative — but pole position keeps him predicted on the podium. Leclerc's 0.16 is far above a normal P19 because his qualifying was better than his grid slot (B3). The function never fails on missing values and uses only pre-race information.

11. Known limitations

- **Qualifying-coverage drift:** no qualifying data before 1994 and only 6–59 % of races in 1996–2002; there qualifying equals the grid, so `grid_penalty_delta` is 0 and the teammate duel compares grid slots.
- **Placeholder values by era:** lap position (< 1996), fastest-lap rank (< 2004) and pit time (< 2011) are fill values; they weaken all-season correlations and create false links between features.
- **Era sensitivity:** reliability, field sizes (podium rate 8–18 %), points systems and front-row conversion changed over time; performance varies by season (PR-AUC 0.65 in 2024 vs 0.76 in 2023), and 2022 introduced new technical rules.
- **Missing information:** weather, tyre strategy, safety cars, in-race reliability and upgrades are not in the data — the main reasons behind both missed podiums and false alarms.
- **Small validation set** (39 races): validation differences below ~0.02 PR-AUC are within noise (bootstrap intervals).
- **Shared drives (1950s):** a few races credit more than three drivers with a podium; **explanations are not causal**.

12. Conclusion

A leakage-safe pipeline turns the 14 relational tables into a validated driver-race table. Starting position is the indispensable signal (ablation, permutation importance, SHAP all agree); recent form, team strength and standings each overlap but together add about 0.07–0.08 PR-AUC. All required models beat the

baselines, and the bonus race-aware ensemble is the best overall (test PR-AUC 0.717, ROC-AUC 0.939, F1 0.683). The remaining errors are driven by race events that no pre-race feature can observe.

Deliverables map

Requirement	Notebook section	Report
EDA before/after cleaning, sentinels, typing, grain, primary/foreign keys	2–4	2
Data-engineering questions with plots (grain, missingness, cutoff)	5, 7.6	3
Feature selection with evidence; leakage-safe form; effect of preprocessing	6, 7	4
Split justification; metrics on train/validation/test; primary/secondary metric	8–12	5, 6
≥ 2 statistical models + shallow NN, baseline, curves, limitations	11–12	6
Feature-group ablation; ≥ 2 preprocessing-order experiments	13–14, 15.6	7
Permutation importance, SHAP global, LIME + local SHAP, non-causal statement	15	8
Inference function on complete and partially-missing records	18	10
Known limitations	7.9	11